

SESSION 1 · SCALING MEDIA SERVICES · SPEAKER SLIDES

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Future Media Townhall

**Media Delivery: Scaling
Media Services**

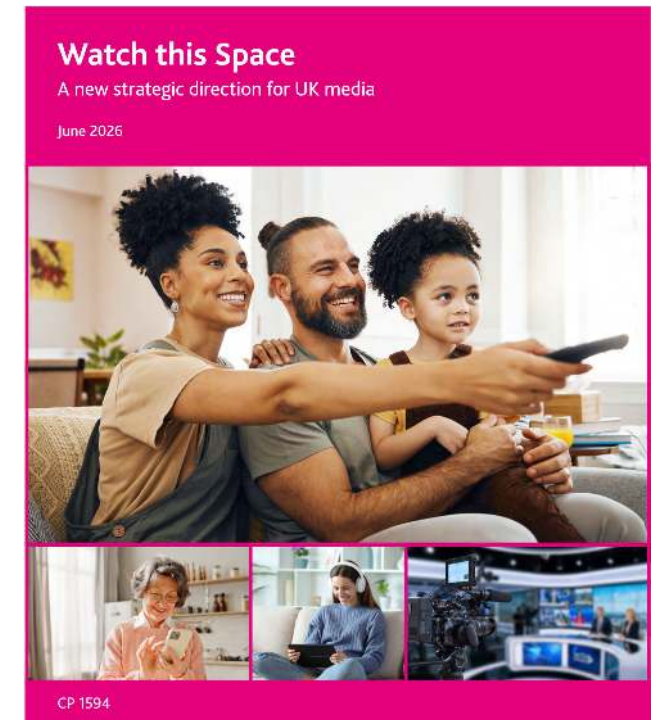
12th Sept 2026 Phil Layton

BBC
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Achieving Scale

UK Planning for an all IP Future

- **June 2026 The UK Government publishes proposals for the closure of DTT/Freeview setting out options of 2034 or 2044 and opens a consultation**
- **Sept 2026 BBC publishes response to the proposals supporting IP transition in 2034**
- **The Connection Project, Digital Television Group, Everyone TV supported by UK public service broadcasters collaborating on producing a proposed transition plan**
- **The UK Government (DCMS) will make the final decision**



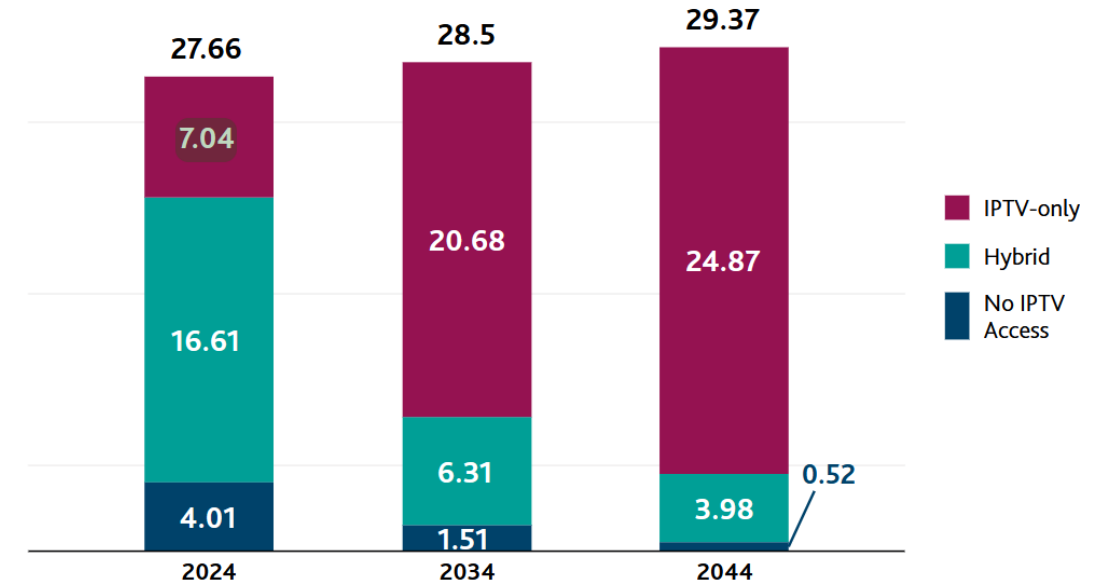
We are already IP

Audience Led Transition

- **Already 7.04m IP only households**
- **By 2029/2030 half UK households will be IP only**
- **This is an audience led transition – different from analogue to digital TV switchover**
- **The 2026 FIFA World Cup - BBC**
 - Peak 6.5m concurrent live streams
 - 1.8m of which were UHD
 - > 60 Tbit/s shipped
- **Estimate 25%-30% of scale required for transition**

B B C R&D

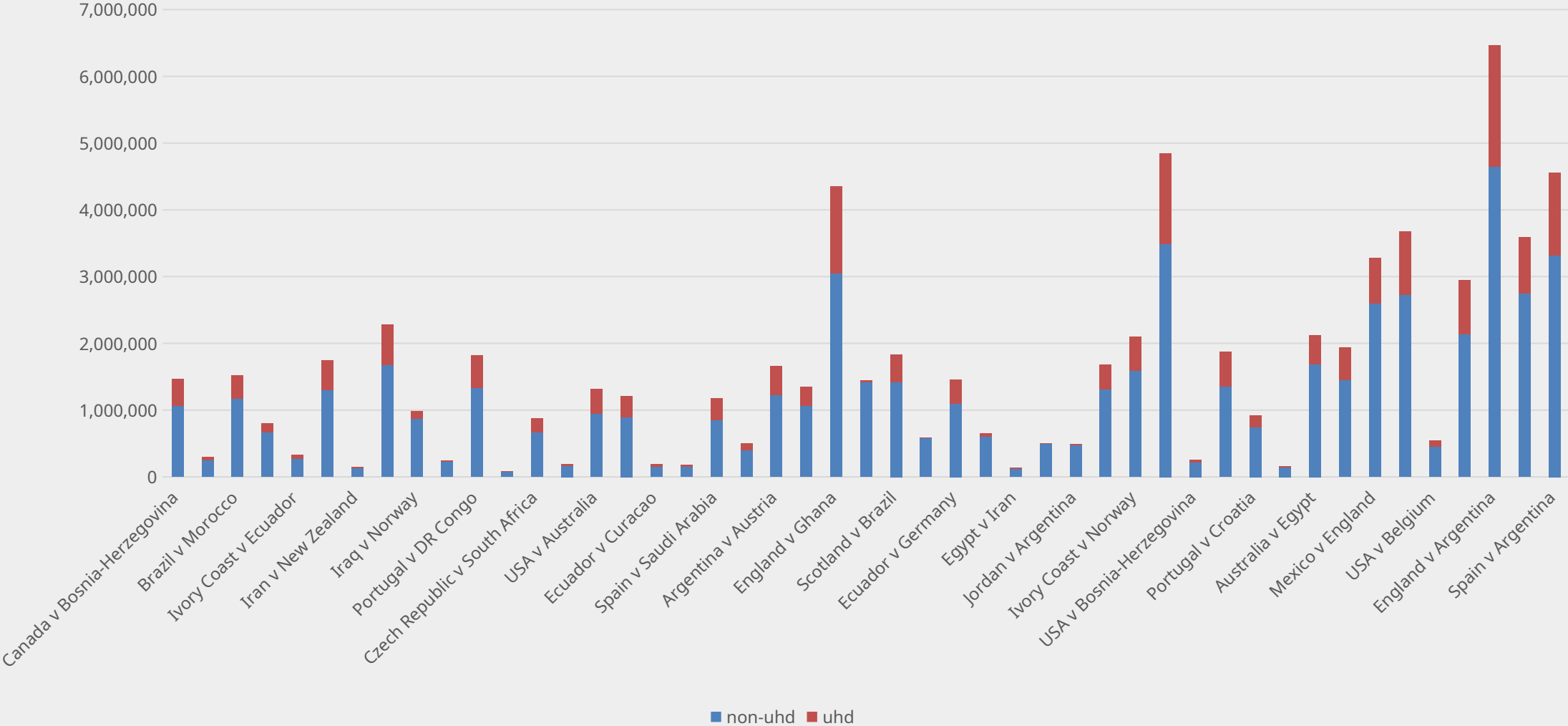
Figure 6: Household IPTV access types (millions)



Source: 3 Reasons, MTM (2024 & 2034 figures); DCMS (2044 figures)

WORLD CUP 2026 MATCHES

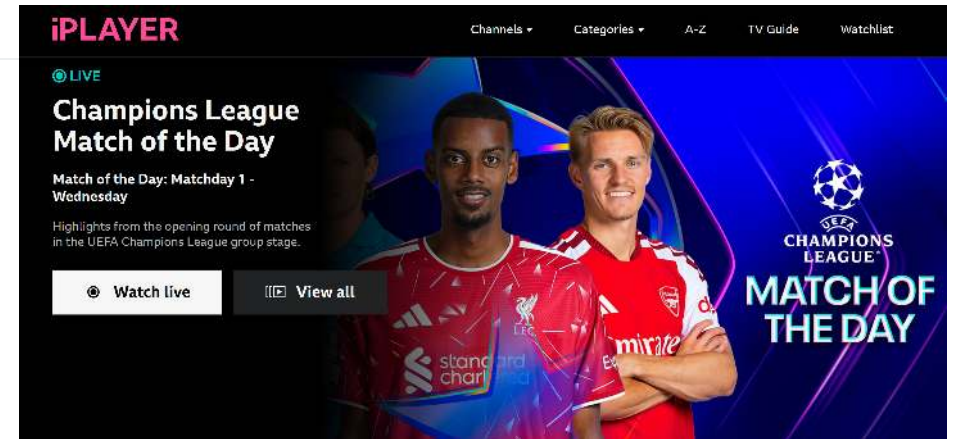
World Cup 2026 Matches (peak streams)



How did BBC achieve this

Delivery to a wide range of devices

- **We operate a multi CDN architecture**
 - Operate our own sovereign CDN – Bidi
 - Embed in ISPs where possible
 - Multiple commercial CDNs
- **End-to-end telemetry for analysis, CDN failover**
 - Client-side logic
- **Deployed our own live packaging solution**
- **Delivery to TVs (iPlayer and Freely), mobile, PCs, game consoles (iPlaye**
- **Event planning with ISPs, CDNs, ITV**



Assume Eight Years to go....

Global events drive scale

- Preserve universality – critical to public service broadcasting
- Grow scale, reliability and resilience of distribution
- Attract viewers and listeners to internet based services
- Enhance the inclusivity of BBC content through personalisation
- A flexible platform for new content formats
- Beyond linear – viewers interact with services

- Working assumption is no new fundamental technologies
- Scale what we have
- Improve telemetry, particularly for Wi-Fi in home
- Deeper, real time performance, fault analysis
- Major events drive scale
 - 2028, 2032 – UEFA European Football Championship
 - 2030 FIFA World Cup

How BBC R&D is Contributing

Scale

- **Designed and trialled Bidi CDN - now rebuilt/scaled by our Digital Distribution colleagues**
 - Trialled virtualised Bidi with BT
- **Ongoing codec efficiency improvements - UHD bit rates down from 34Mbits/s to 18Mbits/s**
 - Capped VBR – long used in broadcast
 - Contributing to new codec development – VVC and next generation
- **With iPlayer colleagues trialled low latency streaming at scale**
 - Achieved 30k HD concurrent sessions in iPlayer across diverse set of TVs/STBs
 - CMAF, Chunked transfer encoding, two commercial CDNs, bespoke encoding/packaging chain
 - Analysing telemetry still but early results suggest minimal degradation

How BBC R&D is Contributing

Did I really say no new technologies.....

- **Evaluation and standardisation of new client technologies e.g. Web Assembly**
- **Evaluation of low earth orbit satellite for live streaming**
 - Already a clear cohort of users
- **Contribution to standards e.g. 3GPP and with 5G-MAG**
 - Consumption of video and audio content in cars, mobiles
- **Collaborative project with Universities of Surrey and Lancaster - AI4ME**
 - Moving from personalised recommendations to personalised content
 - Novel distribution architecture

Thank You

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